Day 097 | Unit 07 | Teacher plan

Understand and Use Percent

Use a Percent Model in Context

Standards: 6.RP.A.3c

Learning target: I can find the part when I know the percent and whole.

enVision Topic 6: Understand and Use Percent | Spectrum companion Chapter 3

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Use part = (percent/100) × whole. You may also find 10%, 5%, or 1% and combine equal pieces. Label the part in the same units as the whole.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 2

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01 / Learn

Use part = (percent/100) × whole. You may also find 10%, 5%, or 1% and combine equal pieces. Label the part in the same units as the whole.

Worked example 1

Find 25% of 180.

1. Use part = (percent/100) × whole.

2. You may also find 10%, 5%, or 1% and combine equal pieces.

3. 25/100 × 180 = 45.

4. State the result in the form requested: 45.

Answer: 45

Worked example 2

A school club has 60 members; 60% take part in a project. Part A: enter the number of participants. Part B: write an equation and explain whether the answer is greater or less than the whole.

1. Use part = (percent/100) × whole.

2. Rename 60% as 3/5.

3. Multiply by 60 to obtain 36.

4. Compare the rate with 1.

Answer: 36 participants; n = (60/100) × 60; less than the whole because the percent is below 100

Day 097 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Find 50% of 220.

02 / Constructed response

Find 10% of 160.

03 / Constructed response

Find 20% of 60.

04 / Constructed response

Find 15% of 200.

05 / Constructed response

Find 5% of 220.

06 / Constructed response

Find 30% of 100.

07 / Constructed response

Find 75% of 160. Then explain a mental-math or fraction strategy that checks your answer.

08 / Numeric entry and equation

A school club has 120 members; 25% take part in a project. Part A: enter the number of participants. Part B: write an equation and explain whether the answer is greater or less than the whole.

09 / Numeric entry and equation

A school club has 40 members; 25% take part in a project. Part A: enter the number of participants. Part B: write an equation and explain whether the answer is greater or less than the whole.

10 / Constructed response

Find 75% of 100. Then explain a mental-math or fraction strategy that checks your answer.

Day 097 | Unit 07 | Answer key

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1. 110

50/100 × 220 = 110.

6.RP.A.3c | Constructed response

2. 16

10/100 × 160 = 16.

6.RP.A.3c | Constructed response

3. 12

20/100 × 60 = 12.

6.RP.A.3c | Constructed response

4. 30

15/100 × 200 = 30.

6.RP.A.3c | Constructed response

5. 11

5/100 × 220 = 11.

6.RP.A.3c | Constructed response

6. 30

30/100 × 100 = 30.

6.RP.A.3c | Constructed response

7. 120

75/100 × 160 = 120.

6.RP.A.3c | Constructed response

8. 30 participants; n = (25/100) × 120; less than the whole because the percent is below 100

Rename 25% as 1/4. Multiply by 120 to obtain 30. Compare the rate with 1.

6.RP.A.3c | Numeric entry and equation

9. 10 participants; n = (25/100) × 40; less than the whole because the percent is below 100

Rename 25% as 1/4. Multiply by 40 to obtain 10. Compare the rate with 1.

6.RP.A.3c | Numeric entry and equation

10. 75

75/100 × 100 = 75.

6.RP.A.3c | Constructed response